

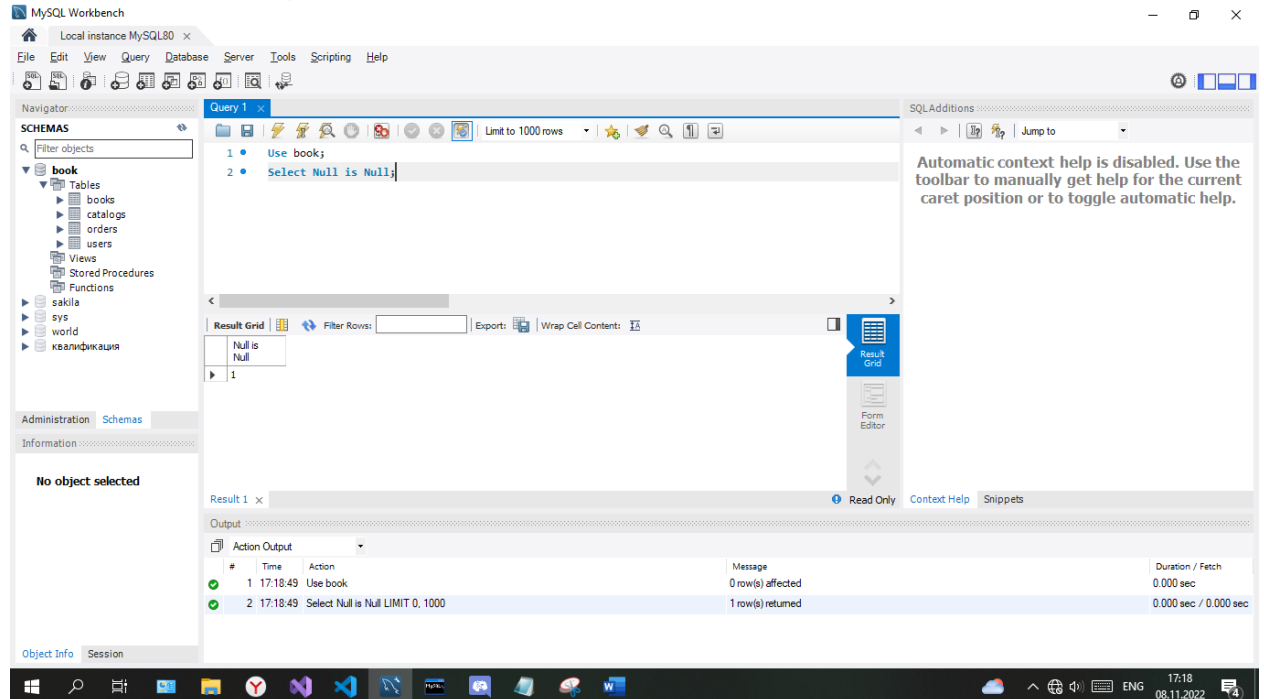
Задание 1

В MySQL Workbench проверьте работоспособность примеров 1-8 на созданной БД «Books».

Пример 1

Use book;

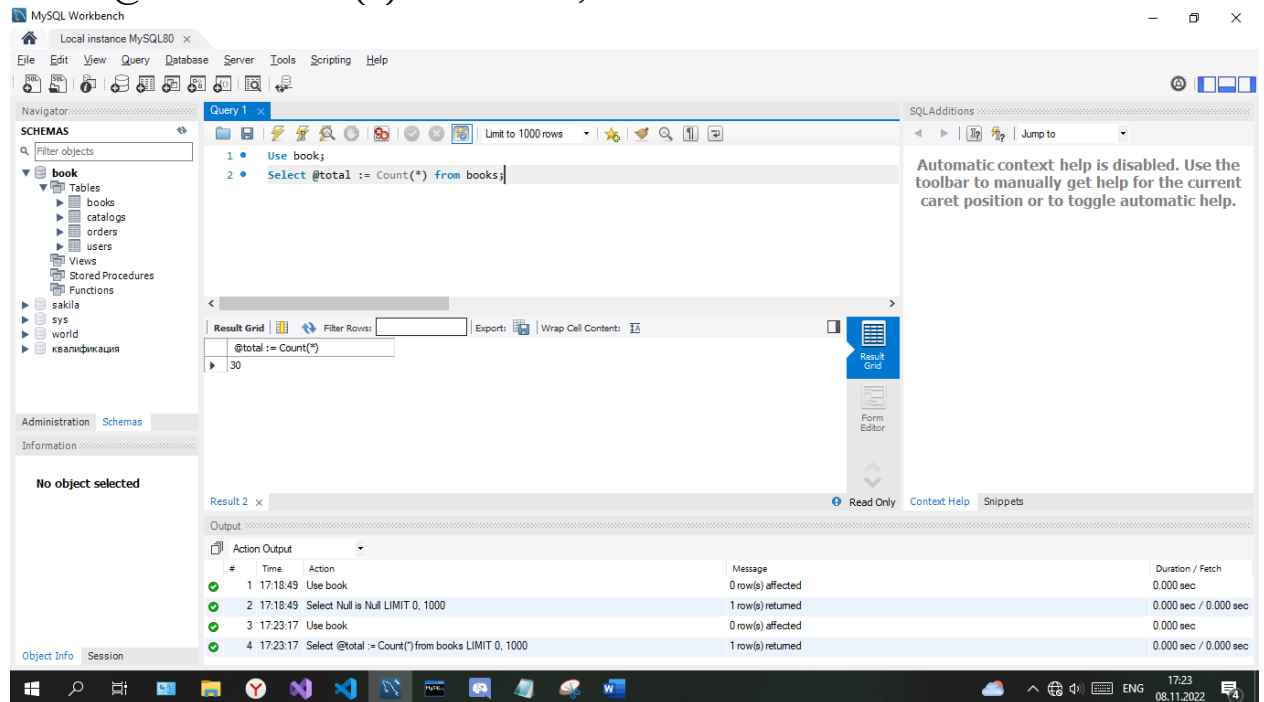
Select Null is Null;



Пример 2

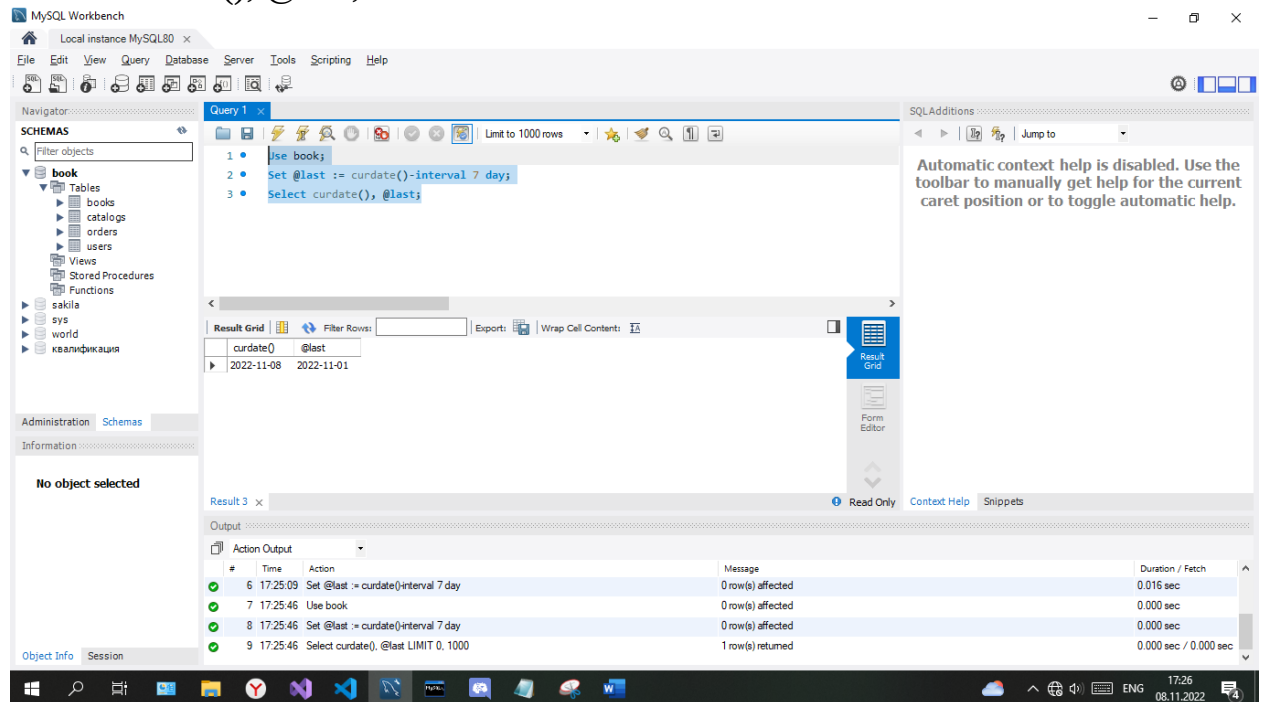
Use book;

Select @total := Count(*) from books;



Пример 3

Use book;
Set @last := curdate()-interval 7 day;
Select curdate(), @last;



MySQL Workbench

Local instance MySQL80

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Filter objects

book

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catalogs

orders

users

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Information

No object selected

Query 1

1 Use book;

2 Set @last := curdate()-interval 7 day;

3 Select curdate(), @last;

Result Grid

curdate() @last

2022-11-08 2022-11-01

Result 3

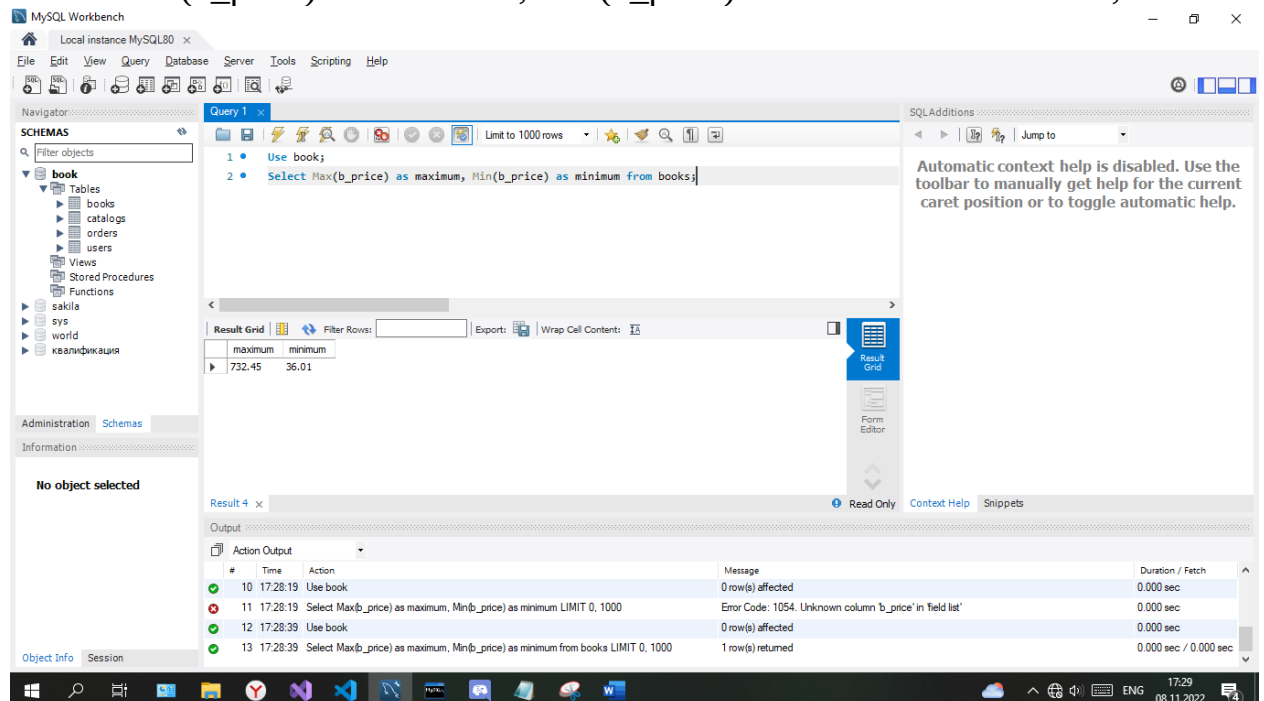
Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	17:25:09	Set @last := curdate()-interval 7 day	0 row(s) affected	0.016 sec
7	17:25:46	Use book	0 row(s) affected	0.000 sec
8	17:25:46	Set @last := curdate()-interval 7 day	0 row(s) affected	0.000 sec
9	17:25:46	Select curdate(), @last LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Пример 4
Use book;
Select Max(b_price) as maximum, Min(b_price) as minimum from books;



MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

book

Tables

books

catalogs

orders

users

Views

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Functions

sakila

sys

world

квалификация

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Information

No object selected

Query 1

1 Use book;

2 Select Max(b_price) as maximum, Min(b_price) as minimum from books;

Result Grid

maximum minimum

732.45 36.01

Result 4

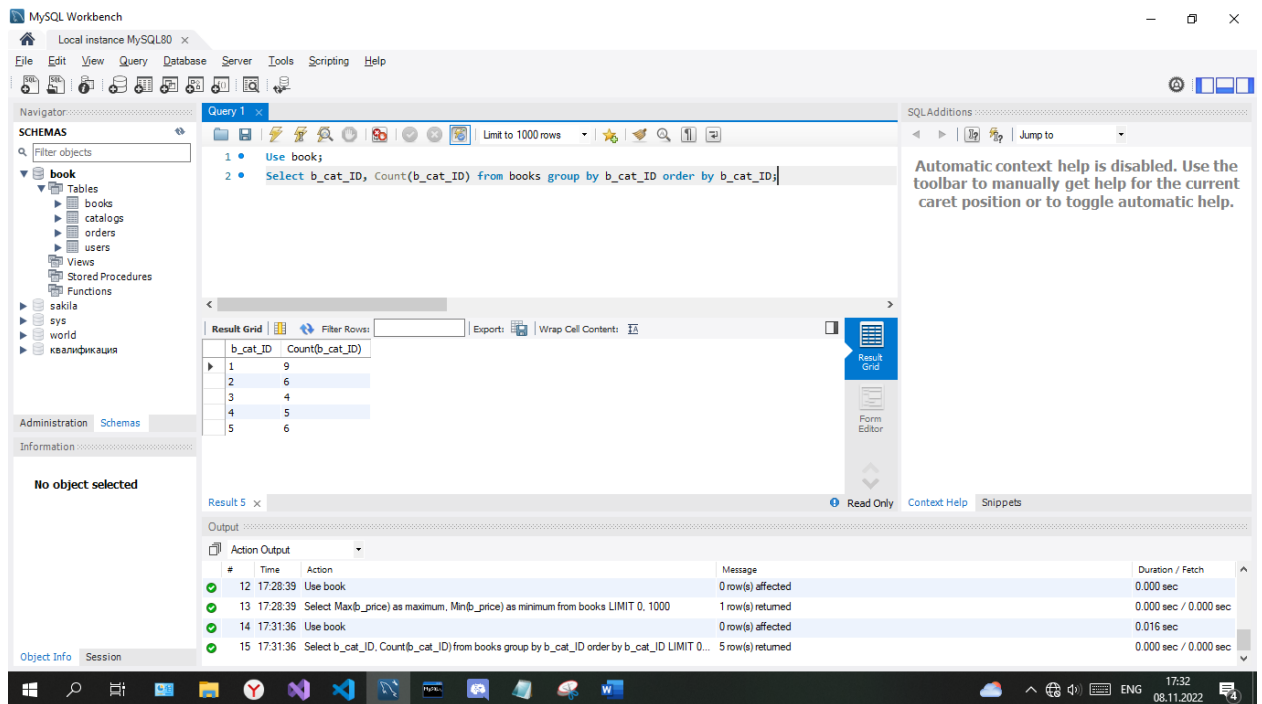
Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	17:28:19	Use book	0 row(s) affected	0.000 sec
11	17:28:19	Select Max(b_price) as maximum, Min(b_price) as minimum LIMIT 0, 1000	Error Code: 1054. Unknown column 'b_price' in field list	0.000 sec
12	17:28:39	Use book	0 row(s) affected	0.000 sec
13	17:28:39	Select Max(b_price) as maximum, Min(b_price) as minimum from books LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

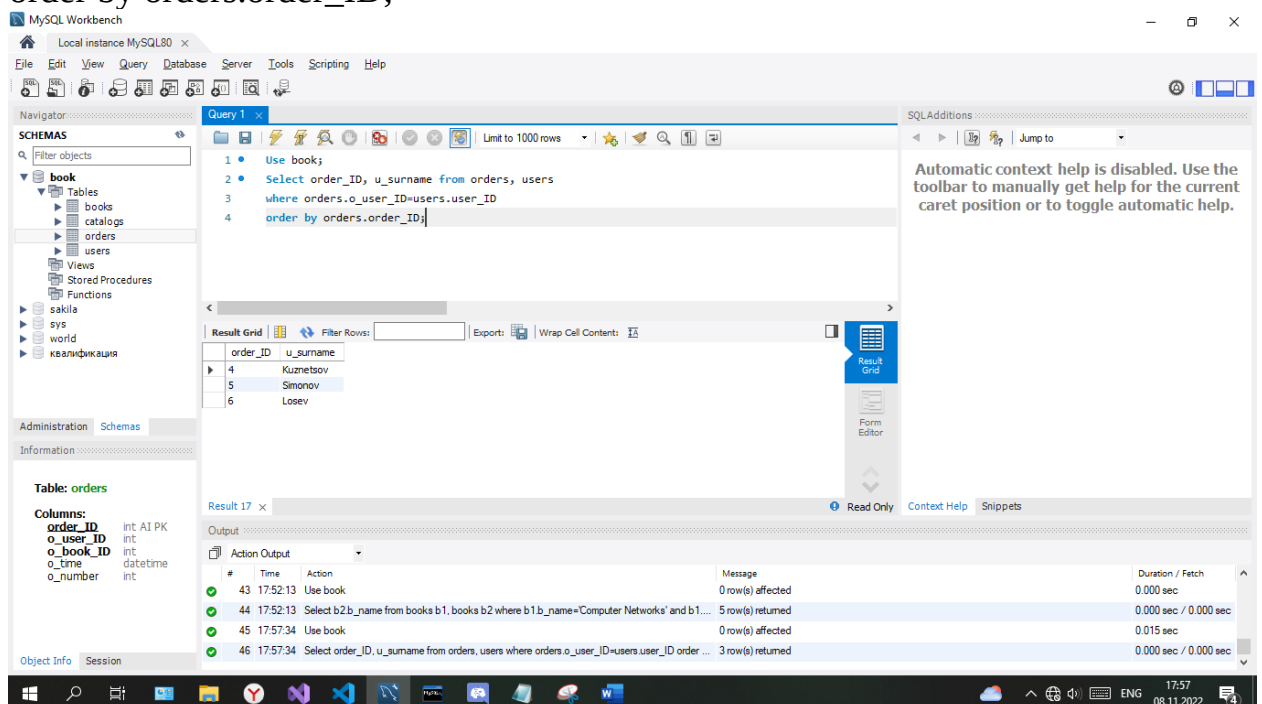
Пример 5
Use book;
Select b_cat_ID, Count(b_cat_ID) from books group by b_cat_ID order by b_cat_ID;



Пример 6

Use book;

Select order_ID, u_surname from orders, users
where orders.o_user_ID=users.user_ID
order by orders.order_ID;

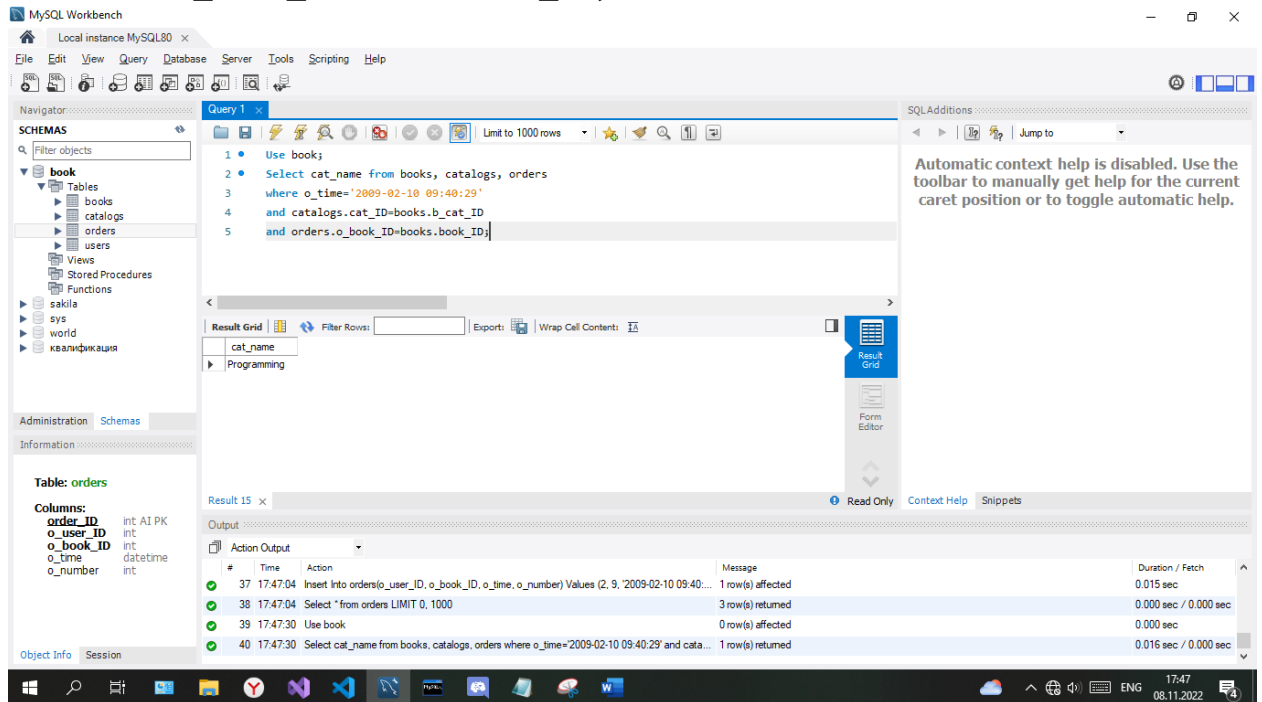


Пример 7

Use book;

Select cat_name from books, catalogs, orders
where o_time='2009-02-10 09:40:29'
and catalogs.cat_ID=books.b_cat_ID

and orders.o_book_ID=books.book_ID;



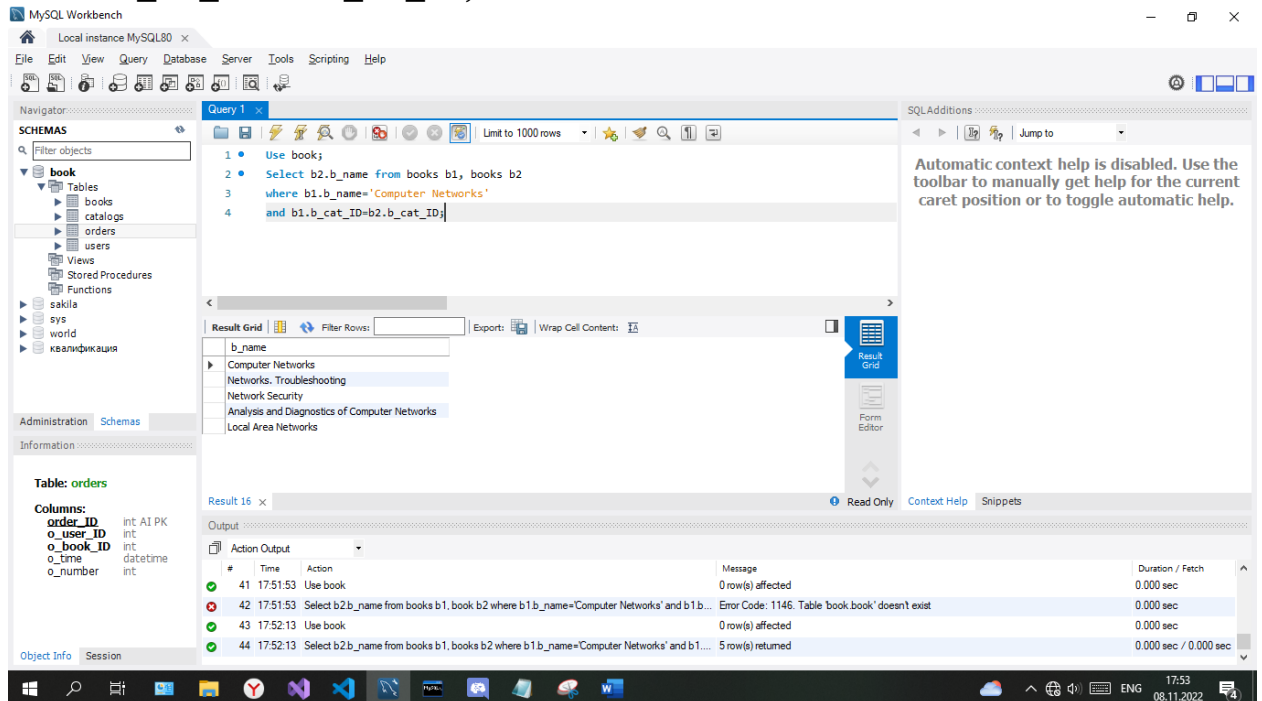
Пример 8

Use book;

Select b2.b_name from books b1, books b2

where b1.b_name='Computer Networks'

and b1.b_cat_ID=b2.b_cat_ID;



Задание 2

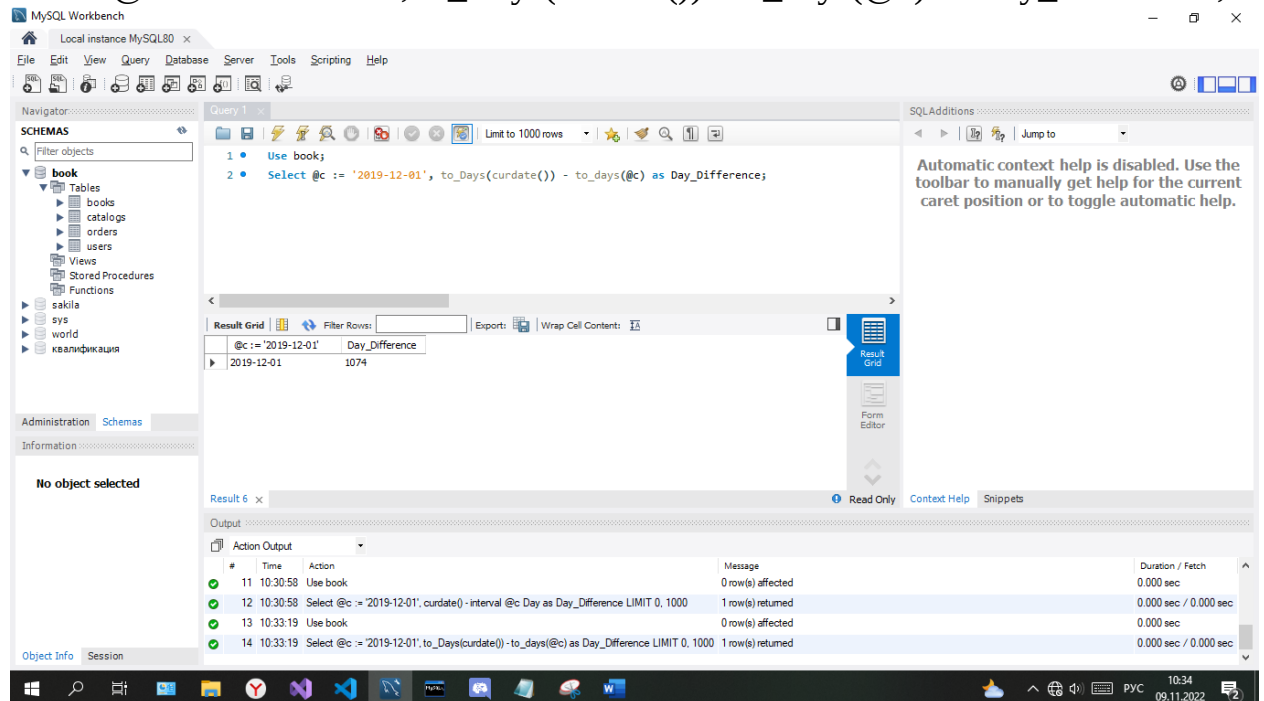
Для примеров 3,4,5,6,7,8 – выполните задание (выделенное зеленым цветом).

Задание 2.3

Способ 1

Use book;

Select @c := '2019-12-01', to_Days(curdate()) - to_days(@c) as Day_Difference;

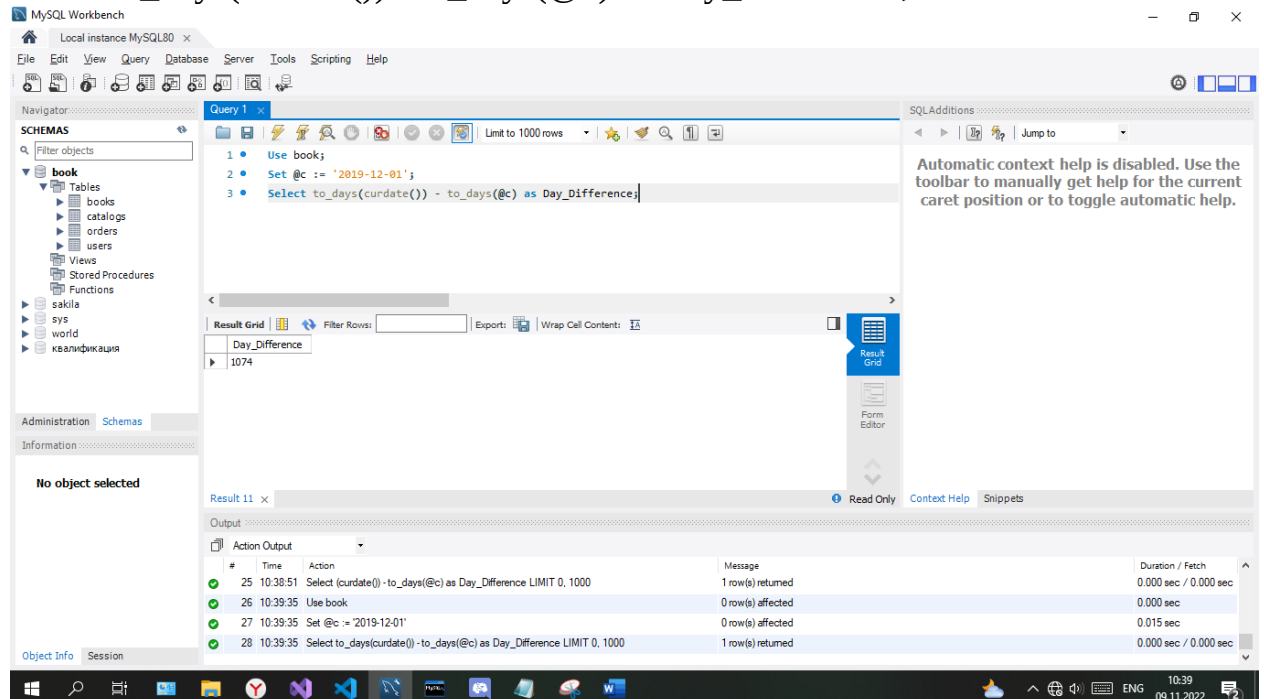


Способ 2

Use book;

Set @c := '2019-12-01';

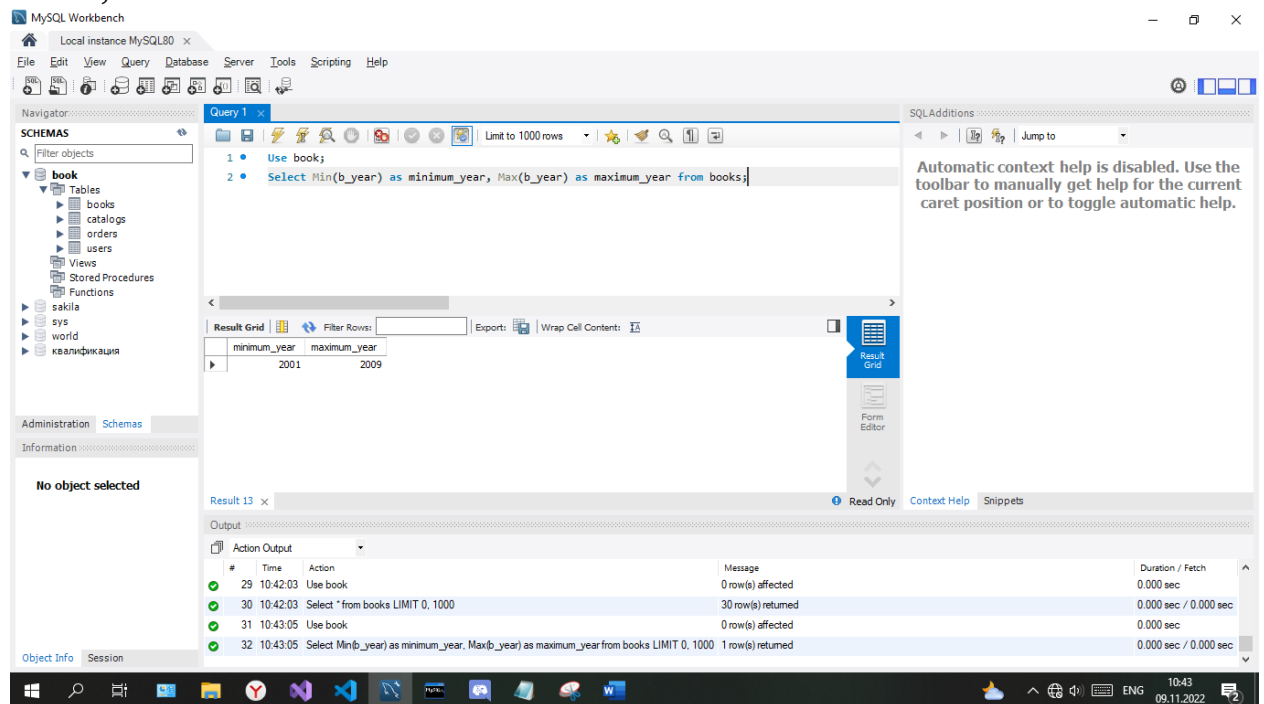
Select to_days(curdate()) - to_days(@c) as Day_Difference;



Задание 2.4

Use book;

Select Min(b_year) as minimum_year, Max(b_year) as maximum_year from books;



The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL query:

```
1 • Use book;
2 • Select Min(b_year) as minimum_year, Max(b_year) as maximum_year from books;
```

The result grid displays the following data:

minimum_year	maximum_year
2001	2009

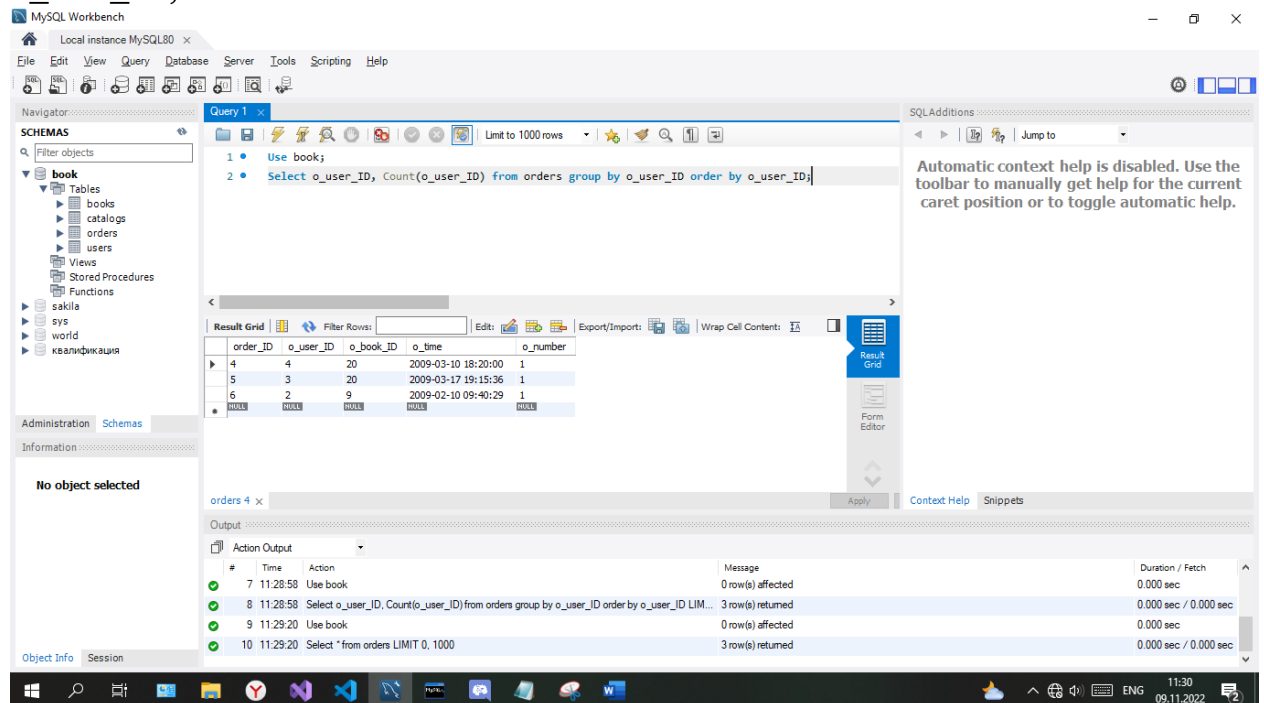
The Action Output pane shows the following messages:

#	Time	Action	Message	Duration / Fetch
29	10:42:03	Use book	0 row(s) affected	0.000 sec
30	10:42:03	Select * from books LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec
31	10:43:05	Use book	0 row(s) affected	0.000 sec
32	10:43:05	Select Min(b_year) as minimum_year, Max(b_year) as maximum_year from books LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Задание 2.5

Use book;

Select o_user_ID, Count(o_user_ID) from orders group by o_user_ID order by o_user_ID;



The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL query:

```
1 • Use book;
2 • Select o_user_ID, Count(o_user_ID) from orders group by o_user_ID order by o_user_ID;
```

The result grid displays the following data:

order_ID	o_user_ID	o_book_ID	o_time	o_number
4	4	20	2009-03-10 18:20:00	1
5	3	20	2009-03-17 19:15:36	1
6	2	9	2009-02-10 09:40:29	1
NULL	NULL	NULL	NULL	NULL

The Action Output pane shows the following messages:

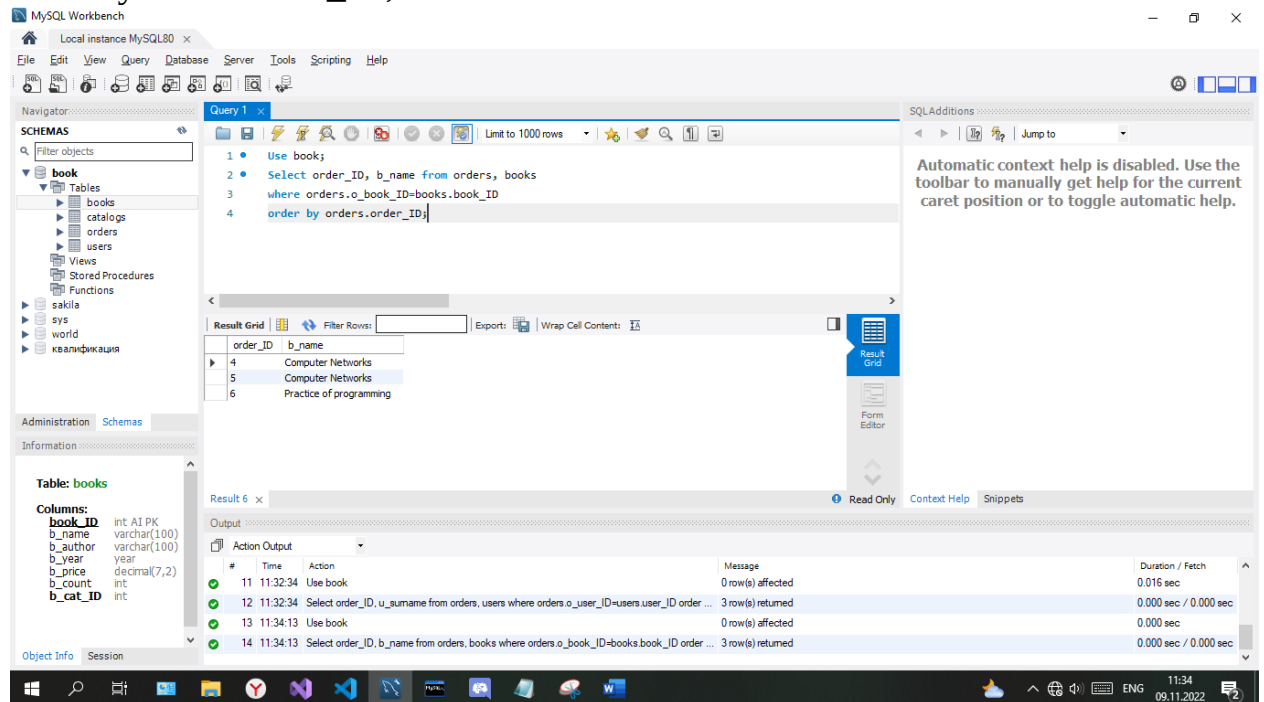
#	Time	Action	Message	Duration / Fetch
7	11:28:58	Use book	0 row(s) affected	0.000 sec
8	11:28:58	Select o_user_ID, Count(o_user_ID) from orders group by o_user_ID order by o_user_ID LIM...	3 row(s) returned	0.000 sec / 0.000 sec
9	11:29:20	Use book	0 row(s) affected	0.000 sec
10	11:29:20	Select * from orders LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Задание 2.6

6.1 Выведите все заказы книги с любым заданным номером книги (o_book_ID)

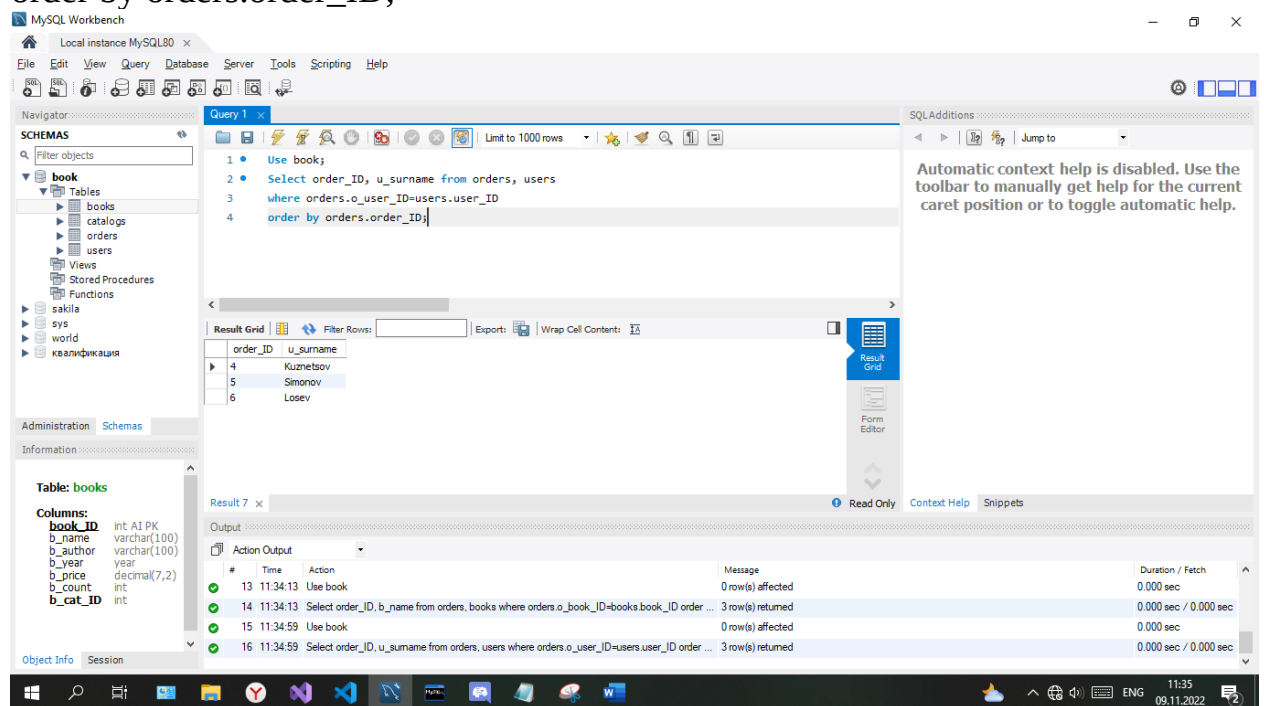
Use book;

Select order_ID, b_name from orders, books
 where orders.o_book_ID=books.book_ID
 order by orders.order_ID;



6.2 Вывести все заказы клиента книги с любым заданным номером клиента (o_user_ID)

Use book;
 Select order_ID, u_surname from orders, users
 where orders.o_user_ID=users.user_ID
 order by orders.order_ID;



6.3 Вывести все заказы, принятые в промежутке любых двух дат (заданных в запросе)

Use book;

Select * from orders

where o_time between '2009-03-01' and curdate()

order by order_ID;

The screenshot shows the MySQL Workbench interface. The 'Query' tab contains the following SQL code:

```
1 Use book;
2 Select * from orders
3 where o_time between '2009-03-01' and curdate()
4 order by order_ID;
```

The 'Result Grid' shows the following data:

order_ID	o_user_ID	o_book_ID	o_time	o_number
4	4	20	2009-03-10 18:20:00	1
5	3	20	2009-03-17 19:15:36	1

The 'Table: orders' information is displayed on the left:

Columns:

- order_ID int: A1 PK
- o_user_ID int
- o_book_ID int
- o_time datetime
- o_number int

The 'Output' pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
23	11:38:41	Use book	0 row(s) affected	0.000 sec
24	11:38:41	Select * from orders where o_time between '2009-03-10' and curdate() order by order_ID LIM...	3 row(s) returned	0.000 sec / 0.000 sec
25	11:38:50	Use book	0 row(s) affected	0.000 sec
26	11:38:50	Select * from orders where o_time between '2009-03-01' and curdate() order by order_ID LIM...	2 row(s) returned	0.000 sec / 0.000 sec

6.4 Вывести все заказы по любому заданному номеру заказа (o_number)

Use book;

Select order_ID, o_number from orders, users

where orders.o_user_ID=users.user_ID

order by orders.order_ID;

The screenshot shows the MySQL Workbench interface. The 'Query' tab contains the following SQL code:

```
1 Use book;
2 Select order_ID, o_number from orders, users
3 where orders.o_user_ID=users.user_ID
4 order by orders.order_ID;
```

The 'Result Grid' shows the following data:

order_ID	o_number
4	1
5	1
6	1

The 'Table: orders' information is displayed on the left:

Columns:

- order_ID int: A1 PK
- o_user_ID int
- o_book_ID int
- o_time datetime
- o_number int

The 'Output' pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
27	11:40:32	Use book	0 row(s) affected	0.000 sec
28	11:40:32	Select order_ID, o_u_surname from orders, users where orders.o_user_ID=users.user_ID order ...	3 row(s) returned	0.000 sec / 0.000 sec
29	11:45:33	Use book	0 row(s) affected	0.000 sec
30	11:45:33	Select order_ID, o_number from orders, users where orders.o_user_ID=users.user_ID order b...	3 row(s) returned	0.000 sec / 0.000 sec

Задание 2.7

7.1 Вывести дату (ы), в которую было сделано максимальное количество заказов.

Use book;

Select @max := Max(o_number) from orders;

Select order_ID, o_time, o_number=@max from orders;

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Local instance MySQL80

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Information

Table: orders

Columns:

order_ID int AI PK

o_user_ID int

o_book_ID int

o_time datetime

o_number int

Object Info Session

Query 1

1 Use book;

2 Select @max := Max(o_number) from orders;

3 Select order_ID, o_time, o_number=@max from orders;

Limit to 1000 rows

Result Grid

order_ID	o_time	o_number=@max
4	2009-03-10 18:20:00	1
5	2009-03-17 19:15:36	1
6	2009-02-10 09:40:29	1
NULL	NULL	NULL

Result 20 orders 21

Output

Action Output

#	Time	Action	Message	Duration / Fetch
46	12:05:31	Set @max = Max(o_number) - 1	Error Code: 1111. Invalid use of group function	0.000 sec
47	12:07:11	Use book	0 row(s) affected	0.000 sec
48	12:07:11	Select @max := Max(o_number) from orders LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
49	12:07:11	Select order_ID, o_time, o_number=@max from orders LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Query Completed

12:07 09.11.2022

7.2 Вывести самую «популярную» книгу.

Use book;

Select * from books order by b_count Limit 1;

MySQL Workbench

Local instance MySQL80

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Information

Table: books

Columns:

book_ID int AI PK

b_name varchar(100)

b_author varchar(100)

b_year year

b_price decimal(7,2)

b_count int

b_cat_ID int

Object Info Session

Query 1

1 Use book;

2 Select * from books order by b_count Limit 1;

Limit to 1000 rows

Result Grid

book_ID	b_name	b_author	b_year	b_price	b_count	b_cat_ID
11	Search on the Internet	Gusev V.S.	2004	91.74	0	2
NULL	NULL	NULL	NULL	NULL	NULL	NULL

books 24

Output

Action Output

#	Time	Action	Message	Duration / Fetch
52	12:09:33	Use book	0 row(s) affected	0.000 sec
53	12:09:33	Select * from books order by b_count desc Limit 1	1 row(s) returned	0.000 sec / 0.000 sec
54	12:09:44	Use book	0 row(s) affected	0.000 sec
55	12:09:44	Select * from books order by b_count Limit 1	1 row(s) returned	0.000 sec / 0.000 sec

Query Completed

12:10 09.11.2022

7.3 Вывести самого «лучшего» клиента (заказавшего максимальное количество книг).

Use book;

Select order_ID, u_Surname, o_number from orders, users

Where orders.o_user_ID=users.user_ID order by o_number Limit 1;

The screenshot shows MySQL Workbench with a query executed. The query is:

```
1 Use book;
2 Select order_ID, u_Surname, o_number from orders, users
3 Where orders.o_user_ID=users.user_ID order by o_number Limit 1;
```

The result grid shows one row:

order_ID	u_Surname	o_number
4	Kuznetsov	1

The bottom panel shows the action output:

#	Time	Action	Message	Duration / Fetch
56	12:12:22	Use book	0 row(s) affected	0.000 sec
57	12:12:22	Select order_ID, u_Surname, o_number from orders, users Where orders.o_user_ID=users.us...	3 row(s) returned	0.000 sec / 0.000 sec
58	12:13:05	Use book	0 row(s) affected	0.000 sec
59	12:13:05	Select order_ID, u_Surname, o_number from orders, users Where orders.o_user_ID=users.us...	1 row(s) returned	0.000 sec / 0.000 sec

Задание 2.8

Use book;

Select o2.o_number from orders o1, orders o2

where o1.o_user_ID=1

and o1.o_user_ID=o2.o_user_ID;

The screenshot shows MySQL Workbench with a query executed. The query is:

```
1 Use book;
2 Select o2.o_number from orders o1, orders o2
3 where o1.o_user_ID=1
4 and o1.o_user_ID=o2.o_user_ID;
```

The result grid shows one row:

o_number
1

The bottom panel shows the action output:

#	Time	Action	Message	Duration / Fetch
67	12:22:10	Use book	0 row(s) affected	0.000 sec
68	12:22:10	Select o2.o_number from orders o1, orders o2 where o1.o_user_ID=1 LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
69	12:22:22	Use book	0 row(s) affected	0.000 sec
70	12:22:22	Select o2.o_number from orders o1, orders o2 where o1.o_user_ID=1 and o1.o_user_ID=o2.o...	0 row(s) returned	0.000 sec / 0.000 sec

Задание 3

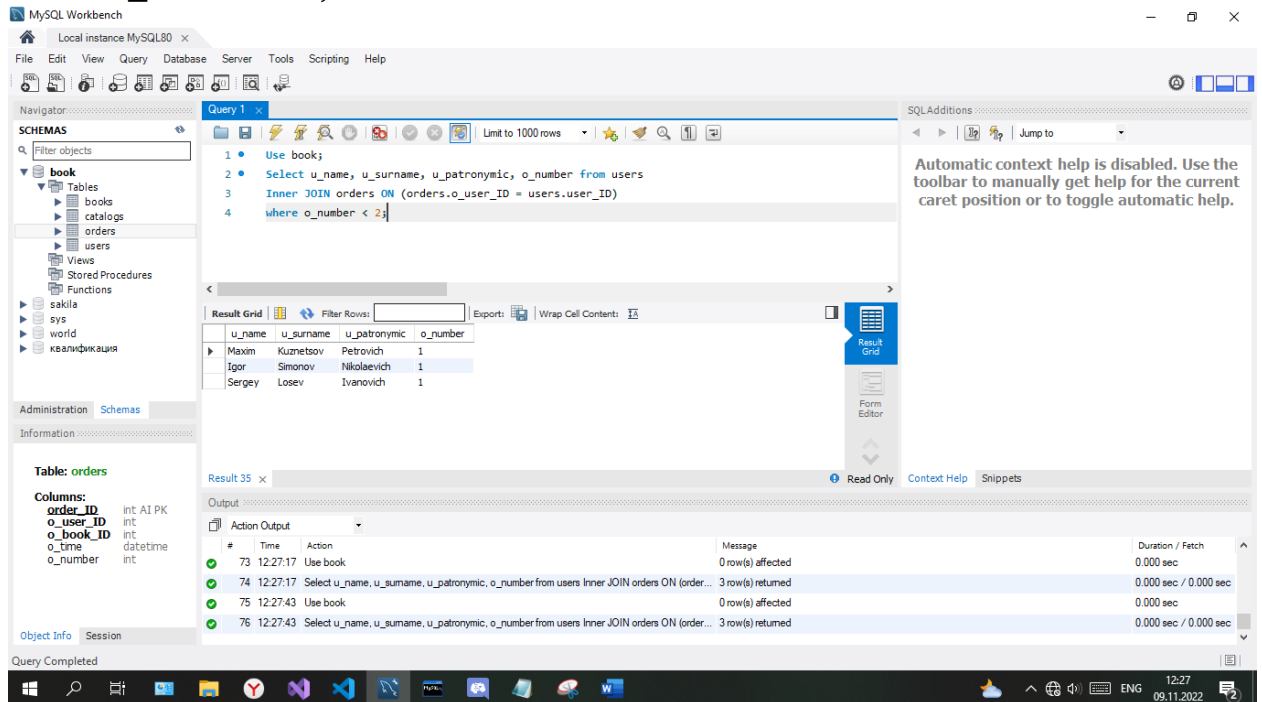
Создайте многотабличный запрос на выборку, который выводит фамилии, имена и отчества покупателей магазина, сделавших менее двух покупок

Use book;

Select u_name, u_surname, u_patronymic, o_number from users

Inner JOIN orders ON (orders.o_user_ID = users.user_ID)

where o_number < 2;



Задание 4

Ответьте на контрольные вопросы:

- 1 Ответьте на вопрос к примеру 1.
Is Null возвращает true/false или 1/0 соответственно. Null является Null, возврат – 1
- 2 Что отличает стандартную функцию MySQL от других конструкций языка?
Наличие круглых скобок
- 3 Сравните результаты следующих выражений на языке MySQL:
1. SELECT 2+3*2; 8
2. SELECT (2+3)*2; 10
- 4 В заданном выражении на языке MySQL укажите что является оператором, что операндами:
1. SELECT @a:=2+3*2;

Операторы: “:=”, “+”, “*”

Операнды: @a, 2, 3

- 5 Укажите номера строчек кода, в которых результат будет неверным:

1. SELECT @a:=2;
2. **SELECT @a=2;**
3. SET @b=2;
4. **SET @b:=2;**

- 6 Что называется сложными запросами? Приведите примеры разных видов сложных запросов.

Запросы, которые обращаются одновременно к нескольким таблицам

Примеры:

```
Select u_name, u_surname, u_patronymic, o_number from users  
Inner JOIN orders ON (orders.o_user_ID = users.user_ID)  
where o_number < 2;
```

```
Select cat_name from books, catalogs, orders  
where o_time='2009-02-10 09:40:29'  
and catalogs.cat_ID=books.b_cat_ID  
and orders.o_book_ID=books.book_ID;
```

- 7 Приведите пример абсолютной ссылки для любой таблицы БД «books»

users.u_surname

- 8 В каком случае применяется самообъединение таблиц? В чем отличие обычного SELECT от SELECT с самообъединением таблиц?

В случае если интересуют связи между строками одной и той же таблицы.